

# NATIONAL UNIVERSITY OF SINGAPORE<br/>SCHOOL OF COMPUTING<br/>CS2106: INTRODUCTION TO OPERATING SYSTEMS<br/>MIDTERM PRACTICE VARIANT B

Time Allowed: 1 Hour<br/>Answer all questions. Questions 1-8 are MCQ; questions 9-12 are multiple-response True/False; questions 13-14 are short-answer questions.

MCQ 1 and 2 are based on this code:

```
int main() {  
    int p = fork();  
    printf("P %d\n", p);  
    if (p == 0) fork();  
    printf("Q\n");  
    return 0;  
}
```

1. How many Q lines are printed?

- a. 1
- b. 2
- c. 3
- d. 4
- e. 5

2. How many processes exist immediately before termination?

- a. 1
- b. 2
- c. 3
- d. 4
- e. 5

3. A running process finishes main() and returns. Which transition occurs?

- a. Running to Blocked
- b. Running to Ready
- c. Running to Terminated
- d. Running to New
- e. Ready to Running

4. A running process spins in a busy-wait loop. Which is most accurate?

- a. It necessarily becomes Blocked.
- b. It remains Running while it tests the condition.
- c. It becomes Terminated.
- d. It enters the New state.
- e. It releases every semaphore.

5. Which setup permits priority inversion?

- a. High and low tasks share a lock, and a medium task consumes CPU.
- b. High and medium tasks share no resources.
- c. Only one low task exists and it never acquires a lock.
- d. All tasks have equal priority.
- e. The high task terminates before requesting anything.

6. Which statement about microkernels is most accurate?

- a. Device drivers must be part of the kernel.
- b. Isolating services in user space can improve fault containment.
- c. IPC is unnecessary.
- d. They cannot support filesystems.
- e. They always outperform monolithic kernels.

7. A parent has 150 ms of CPU work before fork(). After fork(), the parent needs 100 ms and the child needs 50 ms. Under non-preemptive FCFS, with fork CPU-only, who finishes first?

- a. Parent
- b. Child

- c. They finish simultaneously
- d. The process with the lower PID always
- e. Not enough information

8. Which statement about wait(S) and signal(S) is correct?

- a. wait(S) releases a resource.
- b. signal(S) may wake a blocked process.
- c. wait(S) always succeeds immediately.
- d. signal(S) always blocks.
- e. A semaphore cannot have an initial value above 1.

9. Mark each True or False:

- a. A context switch may save and restore the program counter.
- b. Kernel-managed threads can affect scheduling overhead.
- c. Every byte in a process stack must be copied during a context switch.
- d. Storing a fixed-size process state means ten children necessarily cost ten times more to switch.

10. Mark each True or False about system calls:

- a. A system call is an interface to OS services.
- b. A pure arithmetic function necessarily causes a system call.
- c. System calls commonly involve a privilege-level transition.
- d. The OS can execute instructions on behalf of a process during a system call.

11. S starts at 1. Each process executes wait(S); section A; signal(S).

- a. At most one process is in section A at a time.
- b. An arbitrary number of processes may be waiting.
- c. S must become negative when there are waiters.
- d. If every wait has a matching signal, S cannot grow without bound.

12. Mark each True or False about pipes and signals:

- a. A pipe can be used for one-way byte-stream communication.
- b. Closing all write ends allows a reader to observe EOF.
- c. A signal normally carries an entire arbitrary data structure.
- d. A process that forgets to close a duplicated write end may prevent EOF.

13. Exponential averaging uses  $New = 0.6 \text{ Actual} + 0.4 \text{ Old}$ . Ready processes A, B, C have predictions 180 ms, 140 ms, 100 ms. B was blocked and used 60 ms CPU.

- a. Compute B's new prediction.
- b. Which process is selected by non-preemptive SJF?
- c. When should the actual usage be measured?
- d. Explain why copying a running parent's prediction to a new child may disadvantage other ready processes.

14. x starts at 80. Process A performs  $x=x+3$  twice; Process B performs  $x=x-4$  twice. Each individual read-modify-write is non-atomic.

- (a) If both processes use one binary semaphore around both updates, find the possible final values.
- (b) If A and B use separate semaphores, find the possible final values and the minimum/maximum. State the key stale-register interleaving that creates an extreme value.